

zkEVM project 1 & 2

Execution Witness & Guest Program specs/test

April 2026

A decorative light blue triangle is located in the bottom right corner of the slide, pointing towards the top right.

Specs & tests

- Latest release: [zkevm@v0.3.4](#)
 - Support *blocks_test_engine* format (i.e. Hive) [Ignacio]
 - Fixes regarding type definition mismatches with consensus specs [Ignacio]
 - Fix *statelessInputBytes* in fixtures to properly include invalid transactions [Ignacio/Gabriel]
 - Fixtures that use RLP-modifiers should not be used, to be fixed in next release [Ignacio/Gabriel]
- Upcoming:
 - *Schema-id* prefix for multi-fork/future-proof guest input decoding
 - *ChainConfig* inclusion in *StatelessInput*
 - *PublicKeys* proper generation to assist more efficient txs signature verification

EL progress

- Ethrex:
 - [Target running all fixtures in zkevm@v0.3.3](#) [Gaston]
 - [Support canonical SSZ-bytes input in guest program](#) [Ignacio]
 - [Wire missing crypto backend in hash_tree_root for public input](#) [Ben]
 - [Remove rayon](#) and [sync](#) [Ben]
- Nethermind:
 - Progress both in execution witness generation and guest program ([blog post](#)) [Team]
 - Discuss upcoming guest program input design improvements. [Ruben/Ignacio]
- Reth:
 - [Support generating legacy and canonical execution witnesses](#) [Ignacio]
 - [Supporting zkEVM EEST releases](#). [lazovicff]
- Nimbus:
 - Working in running execution witness fixtures. [deme]
- Zilkworm:
 - Experimenting with alternative guest input design that can become more performant. [Som]
 - [Integrating zilkworm as spec-compliant guest program in ere-quests](#) [Developeruche]
- Besu:
 - [zesu supporting v0.3.4](#) [Gabriel/Gary]
 - Found bugs in specs filled tests [Gabriel]

Extra

- Proposed EIP-7709 in ACDE for Hegota. [Ignacio]
 - [Presentation](#) + [Doc with more analysis details](#)
- Execution witness tests surfaced missing tests for BALs surfacing a bug in Nimbus. [Deme, Ignacio, STEEL]
- [Proof encoding scheme proposal](#) [Tau]

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Benchmarking and Metrics



Updates

- Benchmarking:
 - Zisk v0.16.1 run for refreshed EEST benchmarks and mainnet blocks ([dashboards](#))
 - New [Zisk profiling dashboard](#) for easier analysis of any EEST benchmark profiling.
 - Improved [Repricing dashboard](#):
 - Crash message details
 - Unified dataset ux
 - Plot view with proving time per gas cost (linearity)
- EEST:
 - [Found bug and implemented fix in worst_case_miner](#) used in EEST for deep SSTORE benchmarks (bonus: relevant for G* repricings)
 - [Implemented new SLOAD deep branch benchmark](#) in EEST.